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Lab 7

$$400 = \frac{R_f}{R_1}$$

$$R_2 = 2.4k\Omega$$

$$R_1 = 1m\Omega$$

$$C_1 = 1\mu F$$

$$R_5 = 1k\Omega$$

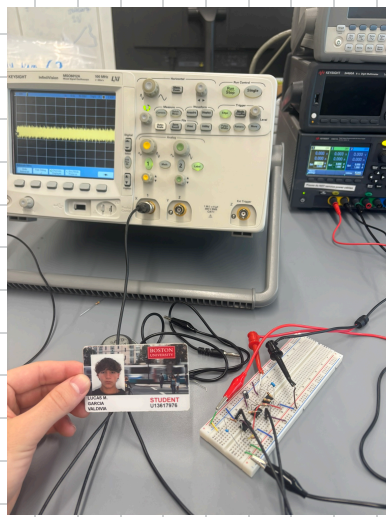
$$R_6 = 1k\Omega$$

$$R_3 = 4.3k\Omega$$

$$R_7 = 100k\Omega$$

$$C_2 = 10\mu F$$

$$C_3 = 1\mu F$$



- First opamp amplifies weak heartbeat signal
- Second opamp uses resistor to further amplify the same signal
- Third Circuit^{1st and 2nd} uses 10nF and 4.3k Ω acts as active filter
- Purpose: clean signal and remove unwanted noise
- Heartbeat sensor converts physical pulses to electrical signal
- two 1k Ω resistor with 3V supply create about 15V
- Capacitors block DC between signal so that only heartbeat signal pulses form
- Resistor control gain

$$V = \frac{3}{2} = 1.5V$$

$$I_{1k\Omega} = 1.5mA$$

$$I_{R_1, R_2, R_3, R_4} = 0mA$$

$$I = \frac{3}{2} = 1.5mA$$

